

HY (GIA) DANG

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RESEARCH INTERESTS

My research focuses on building **reliable and self-improving LLM agents** capable of complex, multi-step reasoning and decision-making. I investigate how agents can effectively use and create tools, learn from experience, and autonomously evolve their capabilities.

My research centers on two directions:

- **Tool-Augmented Agents:** Developing methods for effective tool use and reliable tool construction, improving both agents' tool-use accuracy and the intrinsic reliability of tools in complex, real-world workflows.
- **Self-Improving and Evolving Agents:** Enabling agents to autonomously acquire, refine, and evolve reusable skills and strategies through feedback and experience, improving their adaptability, robustness, and efficiency.

EDUCATION

University of Notre Dame, Notre Dame, IN
Ph.D. Candidate in Computer Science and Engineering
Advisor: Dr. Meng Jiang

August 2022 - Present
GPA: 3.958

Texas Christian University, Fort Worth, TX
B.S. in Computer Science
B.S. in Mathematics

August 2017 - December 2021
GPA: 4.0
Departmental Honors

INDUSTRY EXPERIENCE

Oracle
Applied Scientist Intern (Applied Science Team)

Redwood City, CA
June 2026 – September 2026

- Developed an **agentic AI framework for skill evolution**, enabling LLM agents to construct, refine, and reuse task-specific skills for complex reasoning and decision-making.
- **Mentors:** Dr. Julien Yu, Dr. Mihaela Bornea, Dr. Hiya Roy, Hitesh Patel, Dr. Yinsheng Wang, Dr. Avi Sil (Manager).

Amazon
Applied Scientist Intern (Team Rufus)

Palo Alto, CA
September 2024 - May 2025

- **Project:** Improving the Tool Using and Function Calling Capabilities of LLM(s)
- **Mentors:** Dr. Tianyi Liu, Dr. Zhuofeng Wu, Jingfeng Yang, Dr. Haoming Jiang

RESEARCH PROJECTS

ToolBox Reliability for LLM Agents

May 2025 - present

- Building a community-driven toolbox to improve tool-integrated LLM reliability by addressing both *tool-use accuracy* and *intrinsic tool accuracy*.

Knowledge Augmented & Tool-Use LLM(s)

September 2024 - present

- Working on tool-use capabilities of LLM(s).

Factuality In The Wild & Retrieval-Augmented Generation

April 2024 - September 2024

- Improving automated fact-checking systems, especially for claims in the wild.

Information Retrieval Enhancement Using Text Expansion

August 2022 - April 2024

- Improving diversity in query expansion in document retrieval.

PUBLICATIONS

6. **Hy Dang**, Quang Dao, Meng Jiang. *Open, Reliable, and Collective: A Community-Driven Framework for Tool-Using AI Agents*. In Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP 2026 Demo Track).
5. **Hy Dang**, Yuwen Lu, Jason Spicer, Tamara Kay, Di Yang, Yang Yang, Jay Brockman, Meng Jiang, and Toby Jia-Jun Li. *Uncovering Disparities in Rideshare Drivers' Earning and Work Patterns: A Case Study of Chicago*. Proceedings of the ACM on Human-Computer Interaction (CSCW 2026)
4. **Hy Dang**, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, Haoming Jiang, Tao Yang, Pei Chen, Zhengyang Wang, Helen Wang, Huasheng Li, Bing Yin, Meng Jiang. *Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates*. The Conference on Empirical Methods in Natural Language Processing (EMNLP 2025).
3. Yining Lu, Noah Ziems, **Hy Dang**, and Meng Jiang. *Optimizing Decomposition for Optimal Claim Verification*. Annual Meeting of the Association for Computational Linguistics (ACL 2025).
2. **Hy Dang**, Minh Nguyen, Bo Mei. StTime-Net: Combining both Historical and Textual Factors for Stock Movement Prediction, in Proceedings of *International Conference on Artificial Neural Networks (ICANN)*, Bristol, UK, 2022
1. Quang Truong, Minh Nguyen, **Hy Dang**, Bo Mei. Housing price prediction via improved machine learning techniques, *Procedia Computer Science* 174, 433-442

WORKSHOP PAPERS (* INDICATES EQUAL CONTRIBUTIONS)

2. **Hy Dang***, Bang Nguyen*, Noah Ziems, Meng Jiang. *Embedding Mental Health Discourse for Community Recommendation*. 4th Workshop on Computational Approaches to Discourse, joint with The 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023).
1. Meng Jiang, **Hy Dang**, Lingbo Tong. *A Quantitative Review on Language Model Efficiency Research*. LLM Symposium in conjunction with International Joint Conference on Artificial Intelligence (IJCAI 2023).

PRESENTATIONS

5. **Hy Dang**, Mengxia Yu, Meng Jiang. *Improving Diversity of Query Expansion in Document Retrieval*. Midwest Speech and Language Days (MSLD 2025).
4. **Hy Dang**. *Wound Healing Modeling Using Partial Differential Equations And Deep Learning*. Presentation at National Collegiate Research Conference (NCRC 2022).
3. **Hy Dang**, Luis Mantilla, S. Zhang, Andy Borum. *Bifurcations of an elastic ring with interacting particles*, Student Talk/Poster Session Presentation at the Canadian Undergraduate Mathematics Conference (CUMC 2020).
2. **Hy Dang**. *Wound Healing Modeling Using Partial Differential Equations And Deep Learning*, Poster Presentation at the 3rd Annual Meeting of the SIAM Texas-Louisiana Section, 2020.

1. **Hy Dang.** *Wound Healing Modeling Using Partial Differential Equations And Deep Learning*, Sixteenth Annual Texas Undergraduate Mathematics Conference (TUMC 2021).

AWARDS & FUNDINGS

Outstanding Reviewer

ARR October 2025 Cycle

Tinker Research Grant

Thinking Machines Lab

Award Amount: 5,000 in Tinker credits

OpenAI Researcher Access Program

OpenAI

Award Amount: 10,000 in OpenAI API credits

January 2025 - July 2025

Best Presentation Award

Notre Dame Data Mining Lab

Presentation Title: Expansion is all you need

Spring 2023, Spring 2024

Student Research Symposium Best Poster Award

TCU, Spring 2021

Science and Engineering Research Center Grant

TCU, Fall 2019

Academic Achievement Award

TCU, May 2018

SERVICE

Teaching Assistant at University of Notre Dame: CSE 20110: Discrete Mathematics - Fall 2022, CSE 40171: AI and Society - Spring 2023

Reviewer: CSUR, TKDE 2023, KnowledgeNLP-KDD'23, ICWSM 2024, ICWSM 2025, ARR Review.